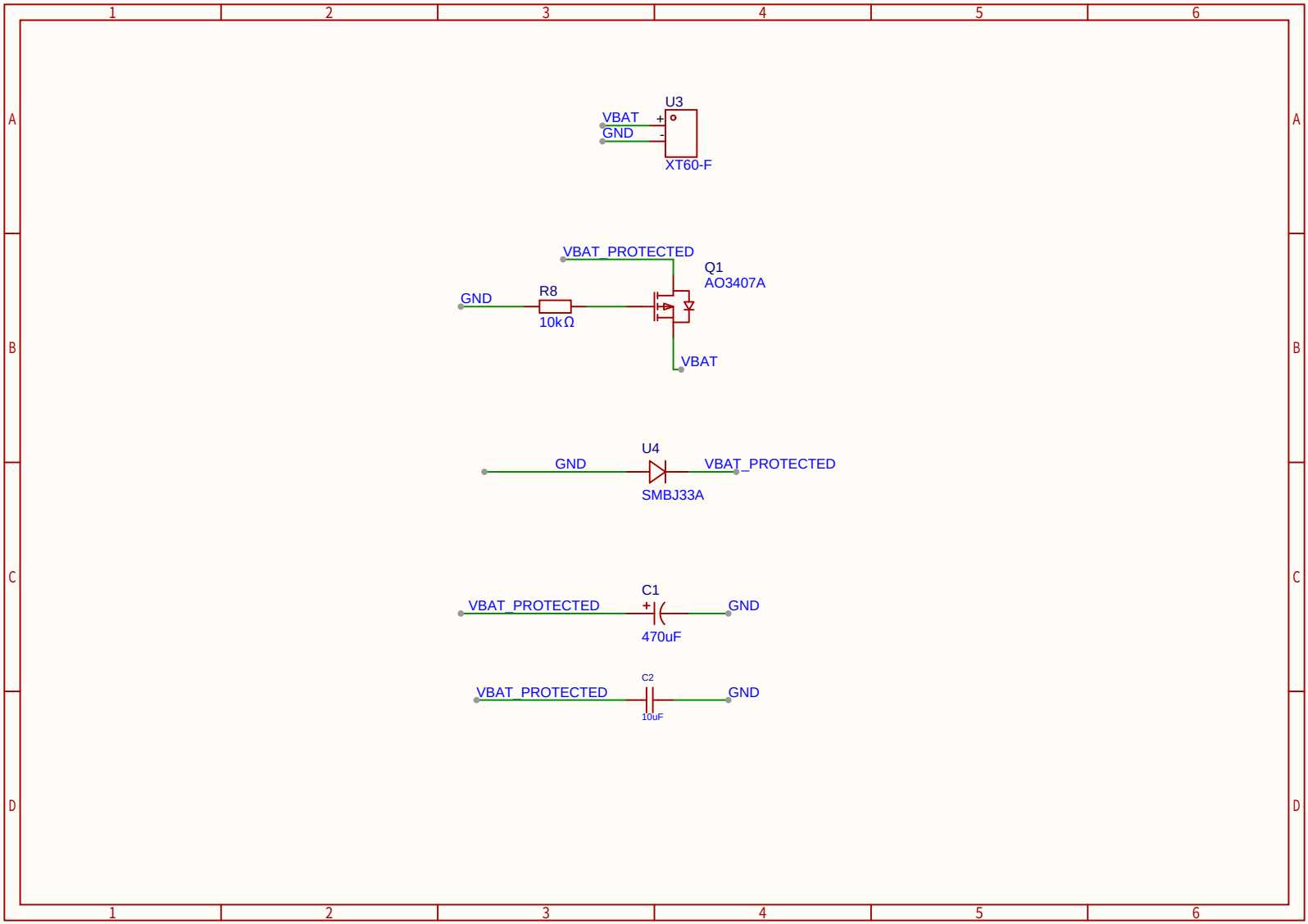
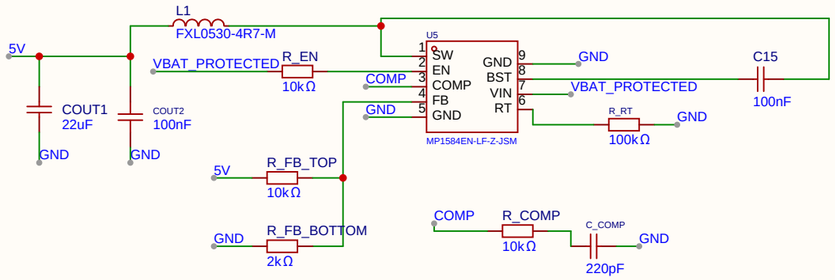
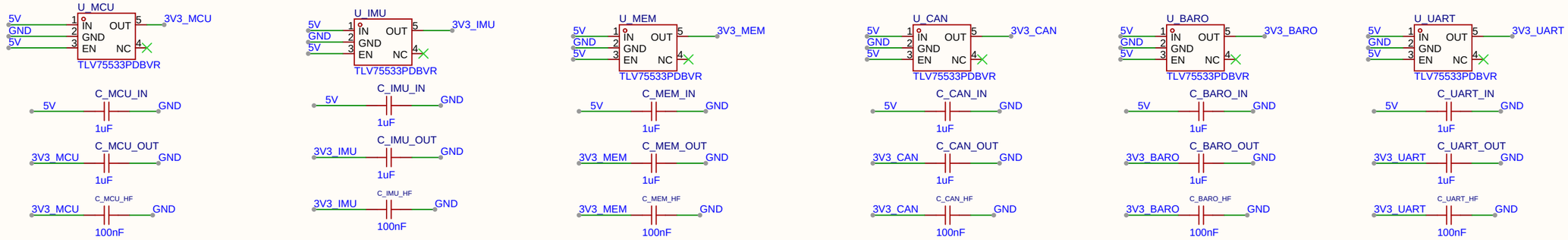
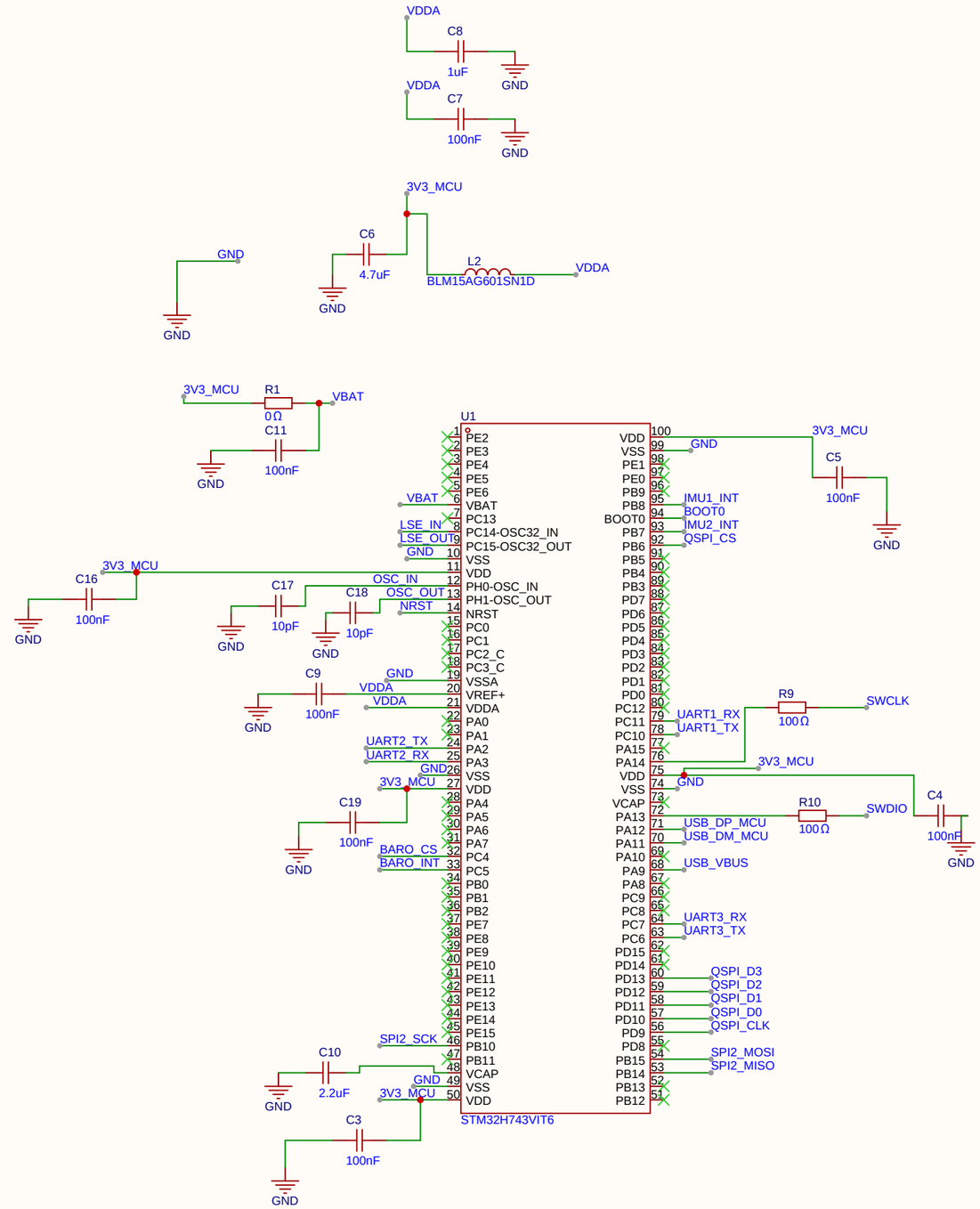
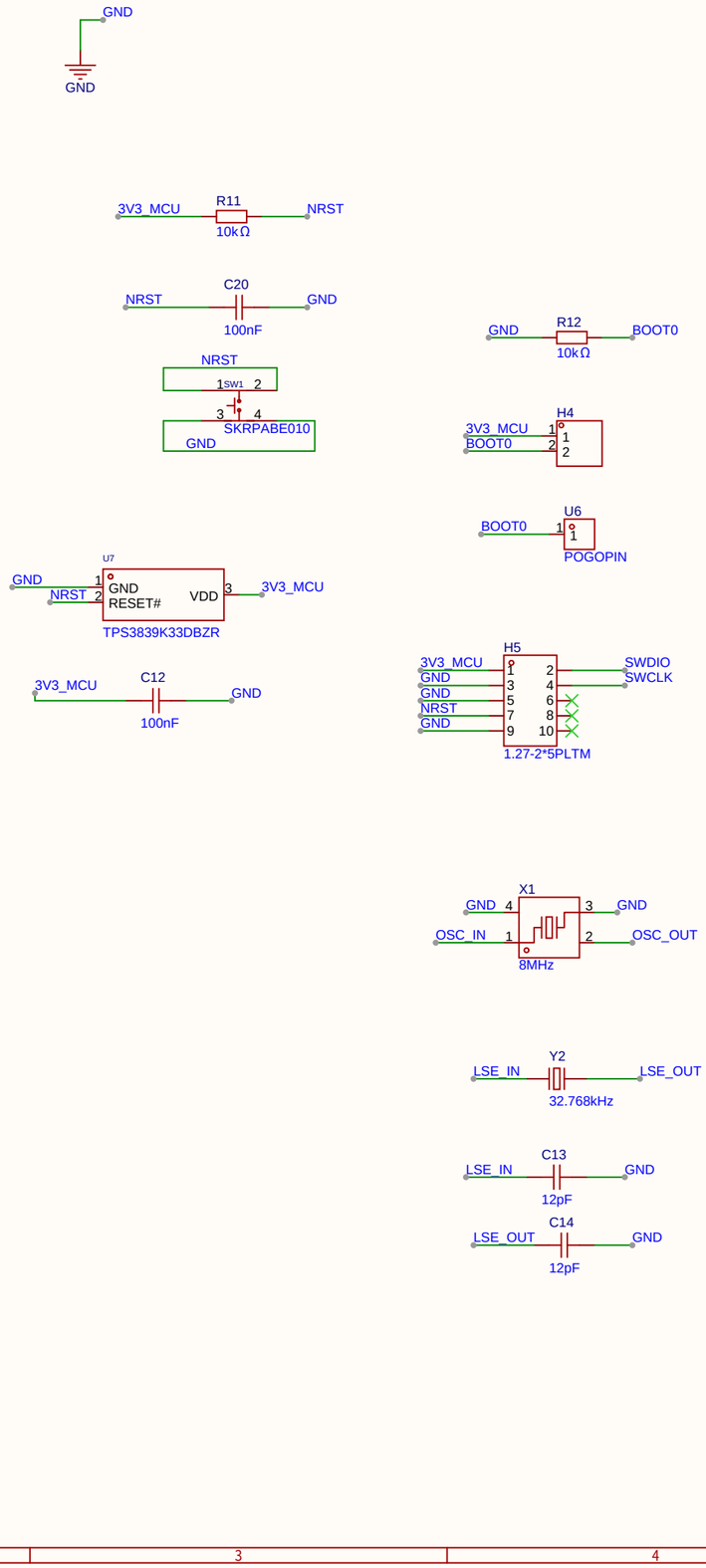
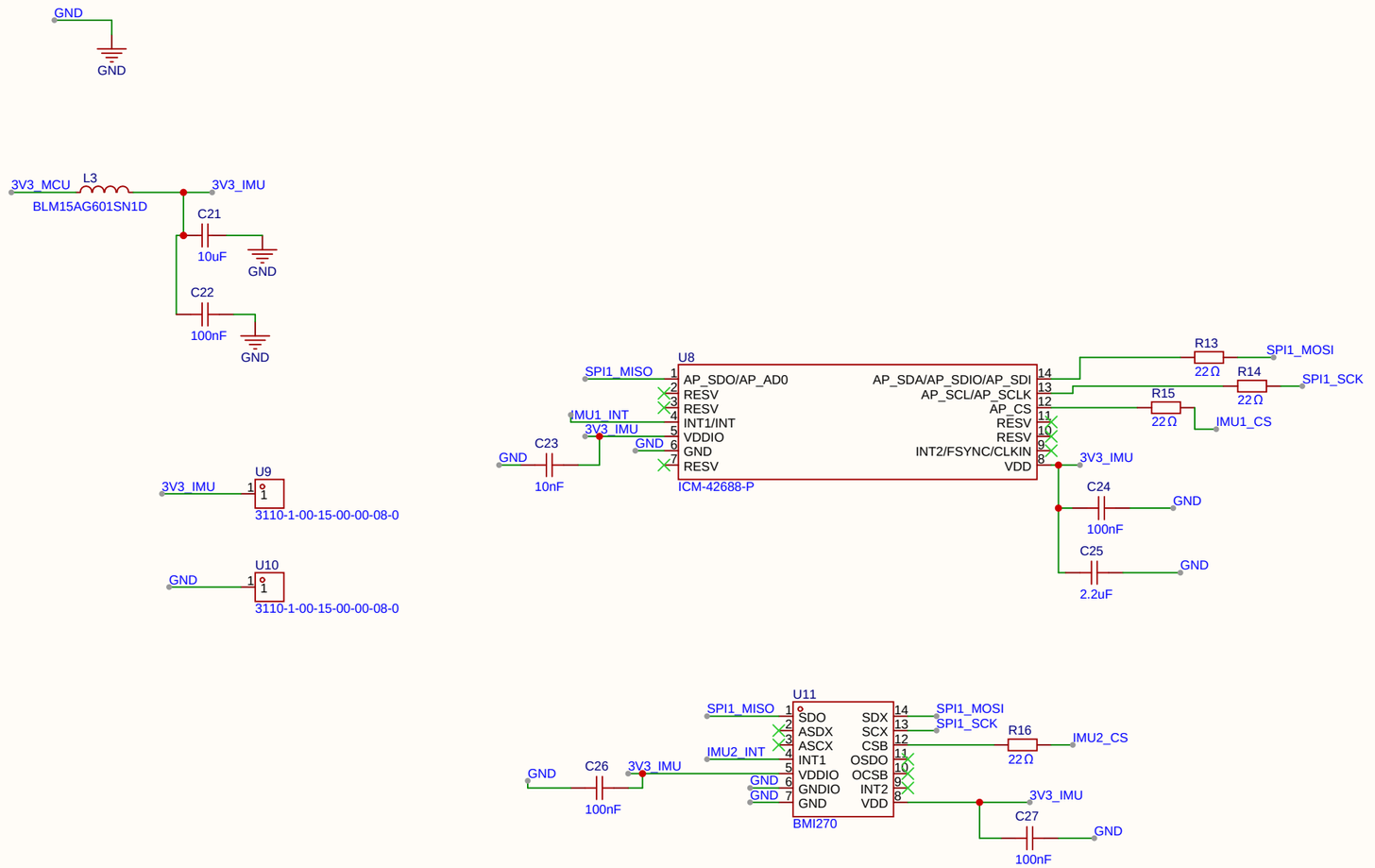

1
2
3
4
5
6
A
B
C
D
U3
VBAT
GND
+
-
XT60-F
VBAT_PROTECTED
Q1
AO3407A
GND
R8
10kΩ
VBAT
U4
GND
VBAT_PROTECTED
SMBJ33A
C1
VBAT_PROTECTED
+
GND
470uF
C2
VBAT_PROTECTED
GND
10uF

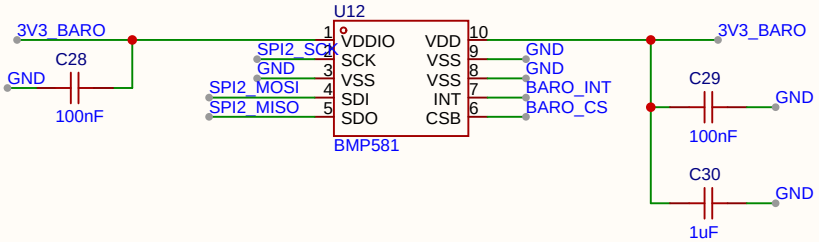

L1
FXL0530-4R7-M
5V
VBAT_PROTECTED
R_EN
10kΩ
COMP
GND
U5
SW
EN
COMP
FB
GND
GND
BST
VIN
RT
MP1584EN-LF-Z-JSM
C15
100nF
R_RT
100kΩ
COUT1
22uF
COUT2
100nF
R_FB_TOP
10kΩ
R_FB_BOTTOM
2kΩ
C_COMP
220pF
R_COMP
10kΩ

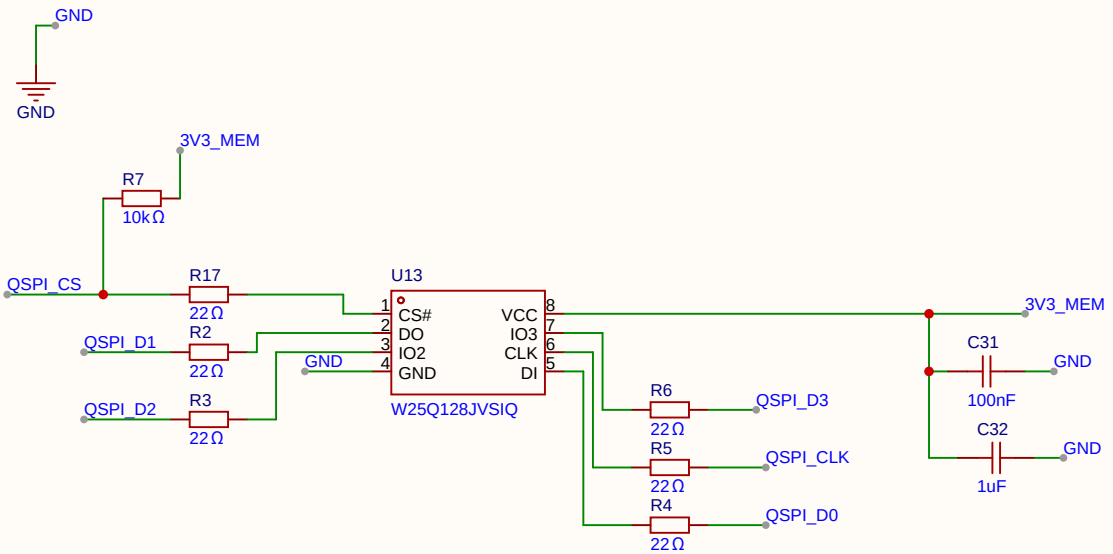

U_MCU
U_IMU
U_MEM
U_CAN
U_BARO
U_UART
IN
OUT
GND
EN
NC
5V
3V3_MCU
3V3_IMU
3V3_MEM
3V3_CAN
3V3_BARO
3V3_UART
TLV75533PDBVR
C_MCU_IN
C_IMU_IN
C_MEM_IN
C_CAN_IN
C_BARO_IN
C_UART_IN
C_MCU_OUT
C_IMU_OUT
C_MEM_OUT
C_CAN_OUT
C_BARO_OUT
C_UART_OUT
1uF
C_MCU_HF
C_IMU_HF
C_MEM_HF
C_CAN_HF
C_BARO_HF
C_UART_HF
100nF

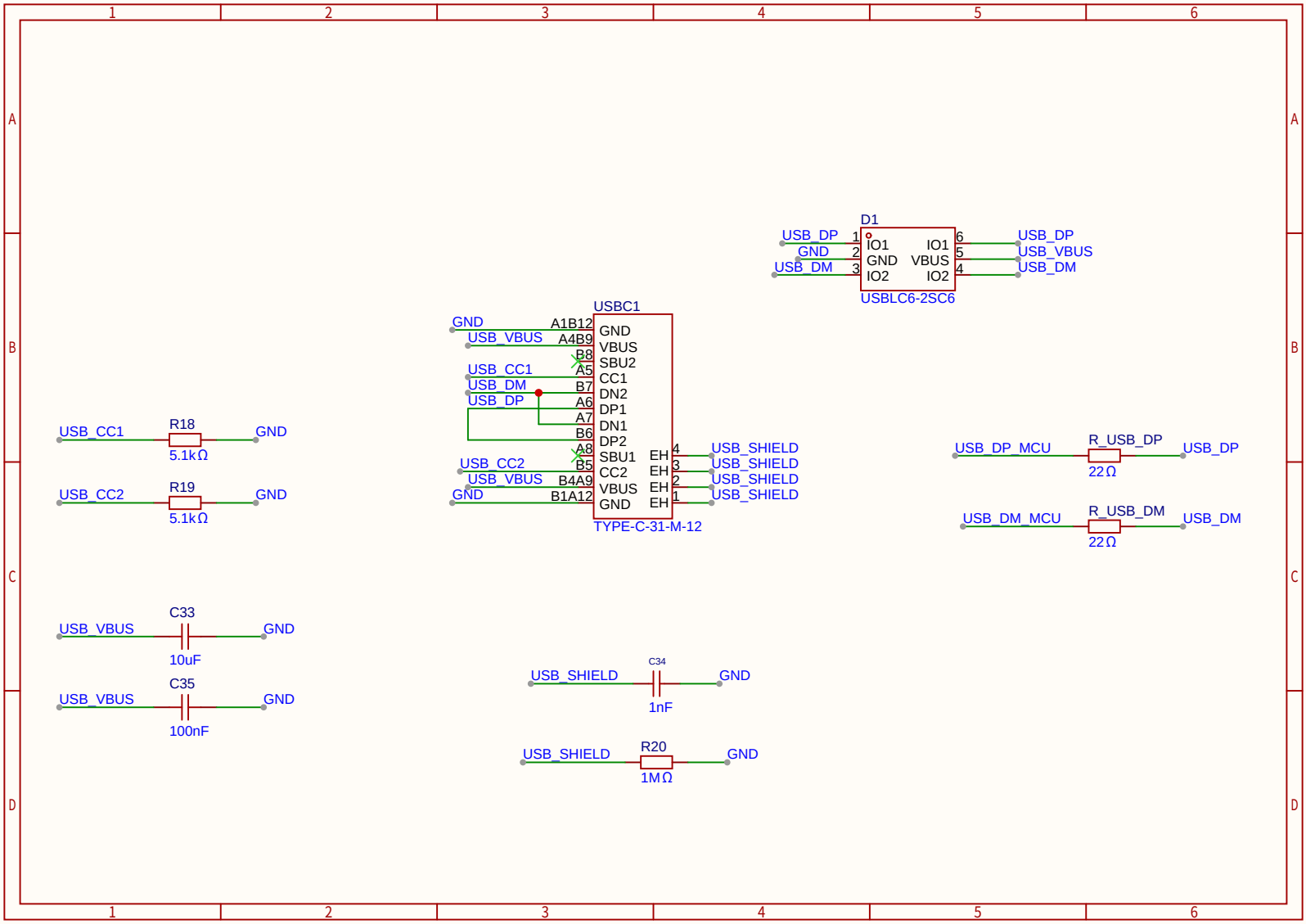

VDDA
C8
1uF
GND
VDDA
C7
100nF
GND
3V3_MCU
C6
4.7uF
L2
BLM15AG601SN1D
VDDA
GND
GND
3V3_MCU
R1
0Ω
VBAT
C11
100nF
GND
U1
PE2
PE3
PE4
PE5
PE6
VBAT
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
VSS
VDD
PH0-OSC_IN
PH1-OSC_OUT
NRST
PC0
PC1
PC2_C
PC3_C
VSSA
VREF+
VDDA
PA0
PA1
PA2
PA3
VSS
VDD
PA4
PA5
PA6
PA7
PC4
PC5
PB0
PB1
PB2
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PB10
PB11
VCAP
VSS
VDD
STM32H743VIT6
VDD
VSS
PE1
PE0
PB9
PB8
BOOT0
PB7
PB6
PB5
PB4
PB3
PD7
PD6
PD5
PD4
PD3
PD2
PD1
PD0
PC12
PC11
PC10
PA15
PA14
VDD
VSS
VCAP
PA13
PA12
PA11
PA10
PA9
PA8
PC9
PC8
PC7
PC6
PD15
PD14
PD13
PD12
PD11
PD10
PD9
PD8
PB15
PB14
PB13
PB12
VBAT
LSE_IN
LSE_OUT
GND
3V3_MCU
C16
100nF
GND
OSC_IN
OSC_OUT
NRST
C17
10pF
C18
10pF
GND
GND
C9
100nF
GND
VDDA
VDDA
UART2_TX
UART2_RX
GND
3V3_MCU
C19
100nF
GND
BARO_CS
BARO_INT
SPI2_SCK
C10
2.2uF
GND
GND
3V3_MCU
C3
100nF
GND
GND
3V3_MCU
C5
100nF
GND
IMU1_INT
BOOT0
IMU2_INT
QSPI_CS
R9
100Ω
SWCLK
UART1_RX
UART1_TX
3V3_MCU
GND
R10
100Ω
SWDIO
C4
100nF
GND
USB_DP_MCU
USB_DM_MCU
USB_VBUS
UART3_RX
UART3_TX
QSPI_D3
QSPI_D2
QSPI_D1
QSPI_D0
QSPI_CLK
SPI2_MOSI
SPI2_MISO

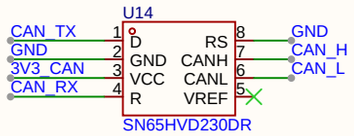

GND
GND
3V3_MCU
R11
10kΩ
NRST
NRST
C20
100nF
GND
NRST
SW1
SKRPABE010
GND
GND
R12
10kΩ
BOOT0
3V3_MCU
BOOT0
H4
BOOT0
U6
POGOPIN
U7
GND
NRST
GND
RESET#
VDD
3V3_MCU
TPS3839K33DBZR
3V3_MCU
C12
100nF
GND
H5
3V3_MCU
GND
GND
NRST
GND
SWDIO
SWCLK
1.27-2*5PLTM
X1
GND
OSC_IN
GND
OSC_OUT
8MHz
Y2
LSE_IN
LSE_OUT
32.768kHz
LSE_IN
C13
12pF
GND
LSE_OUT
C14
12pF
GND

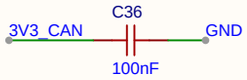

GND
GND
3V3_MCU
L3
BLM15AG601SN1D
3V3_IMU
C21
10uF
GND
C22
100nF
GND
U8
SPI1_MISO
1
AP_SDO/AP_AD0
2
RESV
3
RESV
4
IMU1_INT
INT1/INT
3V3_IMU
5
VDDIO
GND
6
GND
7
RESV
C23
GND
10nF
ICM-42688-P
AP_SDA/AP_SDIO/AP_SDI
14
AP_SCL/AP_SCLK
13
AP_CS
12
RESV
11
RESV
10
INT2/FSYNC/CLKIN
9
VDD
8
R13
22Ω
SPI1_MOSI
R14
22Ω
SPI1_SCK
R15
22Ω
IMU1_CS
3V3_IMU
C24
100nF
GND
C25
2.2uF
GND
U9
3V3_IMU
1
1
3110-1-00-15-00-00-08-0
U10
GND
1
1
3110-1-00-15-00-00-08-0
U11
SPI1_MISO
1
SDO
2
ASDX
3
ASCX
IMU2_INT
4
INT1
5
VDDIO
GND
6
GNDIO
GND
7
GND
SDX
14
SCX
13
CSB
12
OSDO
11
OCSB
10
INT2
9
VDD
8
SPI1_MOSI
SPI1_SCK
R16
22Ω
IMU2_CS
3V3_IMU
C27
100nF
GND
GND
C26
3V3_IMU
100nF
BMI270

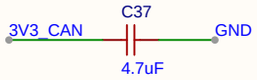

U12
1 VDDIO
2 SCK
3 VSS
4 SDI
5 SDO
VDD 10
VSS 9
VSS 8
INT 7
CSB 6
BMP581
3V3_BARO
C28
GND
100nF
SPI2_SCK
GND
SPI2_MOSI
SPI2_MISO
GND
GND
BARO_INT
BARO_CS
3V3_BARO
C29
100nF
GND
C30
1uF
GND

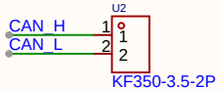

GND
GND
3V3_MEM
R7
10kΩ
QSPI_CS
R17
22Ω
U13
1 CS#
2 DO
3 IO2
4 GND
8 VCC
7 IO3
6 CLK
5 DI
W25Q128JVSIQ
3V3_MEM
C31
100nF
GND
QSPI_D1
R2
22Ω
GND
QSPI_D2
R3
22Ω
R6
22Ω
QSPI_D3
R5
22Ω
QSPI_CLK
R4
22Ω
QSPI_D0
C32
1uF
GND

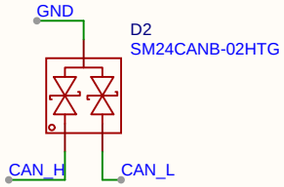

D1
USB_DP
GND
USB_DM
IO1
GND
IO2
VBUS
USB_DP
USB_VBUS
USB_DM
USBLC6-2SC6
USBC1
GND
USB_VBUS
USB_CC1
USB_DM
USB_DP
USB_CC2
USB_VBUS
GND
A1B12
A4B9
B8
A5
B7
A6
A7
B6
A8
B5
B4A9
B1A12
GND
VBUS
SBU2
CC1
DN2
DP1
DN1
DP2
SBU1
CC2
VBUS
GND
EH
USB_SHIELD
TYPE-C-31-M-12
USB_CC1
R18
5.1kΩ
GND
USB_CC2
R19
5.1kΩ
GND
USB_DP_MCU
R_USB_DP
22Ω
USB_DP
USB_DM_MCU
R_USB_DM
22Ω
USB_DM
USB_VBUS
C33
10uF
GND
USB_VBUS
C35
100nF
GND
USB_SHIELD
C34
1nF
GND
USB_SHIELD
R20
1MΩ
GND


U14
CAN_TX
GND
3V3_CAN
CAN_RX
1
2
3
4
D
GND
VCC
R
RS
CANH
CANL
VREF
8
7
6
5
GND
CAN_H
CAN_L
SN65HVD230DR

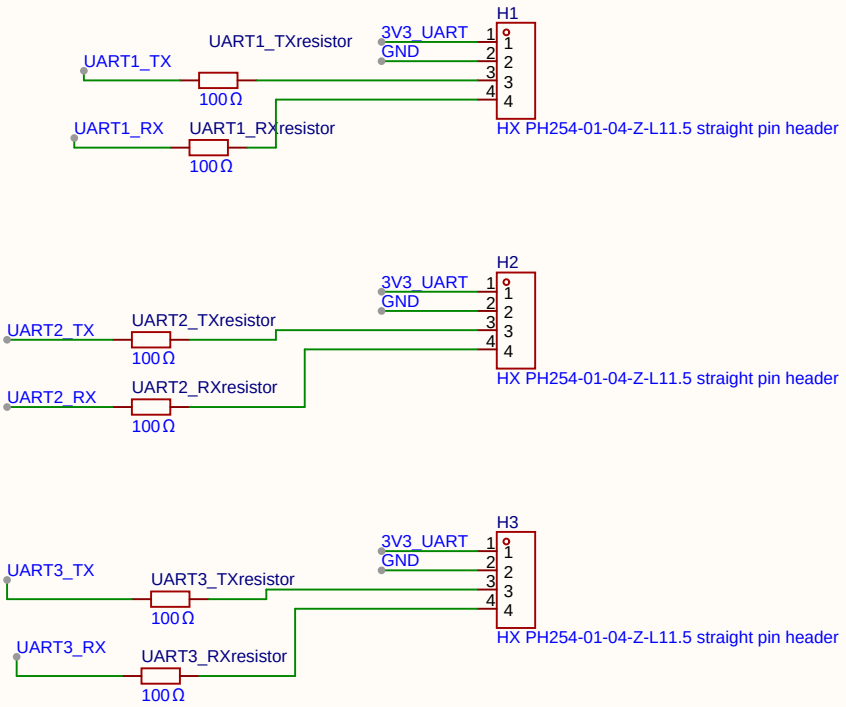

C36
3V3_CAN
GND
100nF

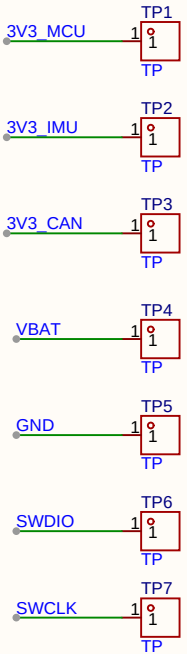

C37
3V3_CAN
GND
4.7uF


U2
CAN_H
CAN_L
1
2
1
2
KF350-3.5-2P


GND
D2
SM24CANB-02HTG
CAN_H
CAN_L

CAN_H
R21
CAN_L
120Ω


H1
UART1_TXresistor
3V3_UART
GND
UART1_TX
100Ω
UART1_RX
UART1_RXresistor
100Ω
HX PH254-01-04-Z-L11.5 straight pin header
H2
3V3_UART
GND
UART2_TX
UART2_TXresistor
100Ω
UART2_RX
UART2_RXresistor
100Ω
HX PH254-01-04-Z-L11.5 straight pin header
H3
3V3_UART
GND
UART3_TX
UART3_TXresistor
100Ω
UART3_RX
UART3_RXresistor
100Ω
HX PH254-01-04-Z-L11.5 straight pin header


3V3_MCU
TP1
1
TP
3V3_IMU
TP2
1
TP
3V3_CAN
TP3
1
TP
VBAT
TP4
1
TP
GND
TP5
1
TP
SWDIO
TP6
1
TP
SWCLK
TP7
1
TP